NTNU
Norwegian University of
Science and Technology

Smart House Energy Modeling with Modelica/Dymola

1. Aim

The aim of this project is to **model the energy use of a typical house in a cold climate** using **Modelica/Dymola** and to evaluate strategies for reducing annual energy consumption. Students will combine real-world weather data with a house model to simulate heating, electricity demand, and the integration of renewable energy and storage systems.

2. Objectives

1. Collect and process **weather data (temperature, solar irradiation, wind speed)** from the national statistics center.
2. Define the **geometry and thermal properties** of a typical house (dimensions, insulation, windows, internal heat capacity).
3. Build a **baseline Dymola+TIL model** of house energy dynamics (heating demand, indoor temperature regulation).
4. Simulate the **annual energy demand** under current conditions.
5. Propose and **model scenarios** for energy reduction and self-sufficiency:
    - Heat pump instead of direct electric heating.
    - Solar panels (PV).
    - Wind turbine (if data supports it).
    - Battery storage.
    - Latent heat storage (phase-change materials).
    - Ventilation with recovery heat exchanger
    - Combination of the above mentioned or other scenarios.
6. Compare scenarios by **annual energy balance, cost, and CO_2 emissions**.

3. Methodology

Step 1: Data collection

Choose location = cold climate region (e.g., −20 °C winter minimum). Download hourly or daily temperature data and wind speed for a full year:

1. Open the website

   Go to https://seklima.met.no/

2. Select stations

   Use the map function on the website to find the meteorological station closest to each of your selected locations. Note the station name.

3. Choose time resolution and weather element

   For Time Resolution, select Hourly.

For Weather Elements, select Air Temperature.

4. Select time period

   Choose the previous year.

   Make sure the downloaded file contains at least 8,760 data points (1 per hour for a year: 24 × 365 = 8,760). If there are fewer data points (e.g., due to missing data or leap years), extend the time period to ensure a full year of hourly data.

5. Download CSV files

If available: solar radiation, wind speed:

1. Access the PVGIS Tool

   Visit https://re.jrc.ec.europa.eu/pvg_tools/en/tools.html

2. Configure the Settings

   Time Resolution: Select Hourly data.

   Optimize Slope and Azimuth: Check the box to optimize panel tilt and orientation. This ensures the system calculates the optimal slope and azimuth angles for maximum solar energy capture.

3. Input Location Details

   Latitude and Longitude: Enter the geographic coordinates of your selected location.

   Period: Choose the previous year (e.g., 2024).

   Mounting Type: Select Free-standing or Roof-mounted, depending on your setup.

4. Select Radiation Components

   Radiation Components: Choose Global Horizontal Irradiation (GHI). This is the total solar radiation received per unit area on a horizontal surface.

   Note: *Some locations may provide irradiance data as three separate components: Gb(i) (direct), Gd(i) (diffuse), and Gr(i) (reflected). In such cases, sum these components to obtain the total GHI:*

$$G(i) = Gb(i) + Gd(i) + Gr(i).$$

5. Download the Data

   After configuring the settings, click on "Download Data".

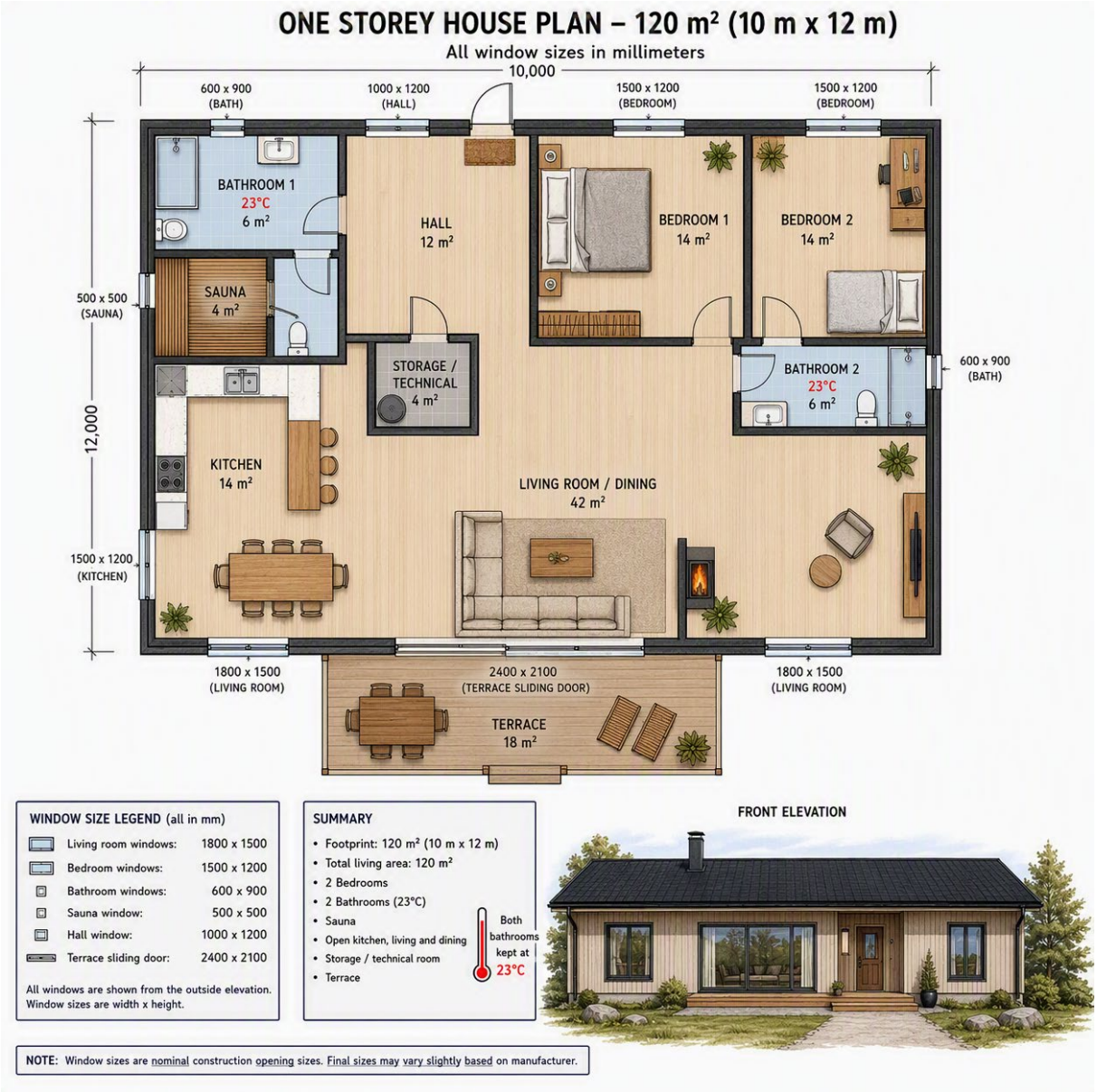
   Choose the CSV format for easy integration with analysis tools.

   Save the file with a descriptive name, such as LocationName_GHI_2024.csv.

Step 2: Baseline house model

For your project work, the following picture contains the layout of the house, which includes window sizes, wall sizes, ground floor, etc.

The house model must follow the basic parameters provided below and must contain of at least 5 different indoor climate zones/rooms, with different load profiles/utilisation.



Basic geometry & volume of house (or use TEK17 standard)

- Heated floor area: 120 m²
- Footprint: 10 m × 12 m = 120 m²
- Number of storeys: 1 (each storey 2.5 m) → total height = 2.5 m
- Heated volume: 120 m² × 2.5 m per storey × 2 storeys = 300 m³
- Roof area: ≈ footprint = 120 m²
- Ground floor area (floor-on-ground): 120 m²

- **U values:**
  - External walls (U): 0.18 W/(m²·K) — (well insulated timber-frame or cavity with ~200 mm insulation)
  - Roof / attic (U): 0.13 W/(m²·K) — good loft insulation (~300 mm mineral wool)
  - Ground floor slab / floor-on-ground (U): 0.15 W/(m²·K) — insulated slab or suspended insulated floor
  - Windows (glazing + frame, average) (U): 1.20 W/(m²·K) — good triple glazing; double glazing would be ~1.6–2.0
  - External door (U): 1.50 W/(m²·K)
- **Ventilation heat loss:**
  - Assume air changes per hour (ACH) = 0.5 (typical for a moderately tight house with some infiltration).
  - The house has an active ventilation system with heat recovery.
- **Thermal mass / internal heat capacity:**
  - Choose a moderate thermal mass representative for typical timber-construction with concrete ground floor + interior furnishings:
  - Specific heat capacity assumption (typical average thermal mass): 150 kJ/(m²·K) (this is a practical modelling parameter).
- **Internal gains (typical averages to include in simulations)**
  - Occupants: 4 persons average produce sensible heat≈ 70 W/person.
  - Appliances / cooking (average): 150–300 W (use 200 W typical)
  - Lighting (modern LED average): 30–80 W (use 50 W)
  - Total internal gains (average): ~425 W (use 300–600 W range depending on occupancy profile)
- **Assume a typical how water demand profile for a one-family house throughout a day/week.**

Set up a class “Room”, which serves as a framework for your simulations. The class “Room” should be able to sufficiently represent a real room, including:

- Heating from (hydronic (water)) floor heating
- Heat loads from the ventilation system per room
- The heat capacitance of the walls
- Heat load profiles of appliances and usage profiles

Step 3: Auxiliary systems:

As you might have noticed, the house consists of a heating and a ventilation system. This should be included in your system model.

The hydronic heating system should include a domestic hot water storage system and an electric boiler to produce warm water. They should be connected in such a way that stratification is ensured.

Step 4: Extensions

The following extensions should be modelled.

- **Heat pump**: replace heater with COP-dependent heating system (TIL library).
- **PV system**: model solar generation using solar irradiation and panel efficiency.
- **Wind system**: approximate with wind speed → power curve.
- **Battery storage**: Modelica.Electrical.Storage.Battery.
- **Latent heat storage**: use a phase-change model for thermal storage (TIL library).

Step 5: Scenario analysis

- Simulate a typical winter week and a typical summer week
- Evaluate amongst others the following metrics:
  - Total heating demand (kWh/week).
  - Share of renewables (%).
  - Grid import/export.
  - Energy cost and CO_2 savings.

4. Expected Deliverables

1. Modelica models of house + energy systems.
2. Simulation results (plots of temperature, demand, PV/wind production, storage levels).
3. Comparison table of scenarios.
4. Report with conclusions: which combination gives best savings and comfort.

Project Report Guidelines – Smart House in Modelica/Dymola

For this project, students should develop their models using **Modelica** in the **Dymola** environment while using the TIL library. Dymola will be used for building component-based models, setting parameters, running simulations, plotting results, and analysing system behaviour. The submitted project folder must include the Dymola-compatible Modelica package that have been developed and used to produce the results. The assignment aims to provide training in **Modelica and energy system modeling**; however, the **main focus in the assessment is the project report and the understanding of the technical concepts**.

- Students should be able to use **Dymola** to build models, run simulations, and visualize results in a clear and readable way.
- The report must be written according to the standards of **scientific report writing**. It should be a coherent text, where **figures and tables** are used to support the presentation of results.
- The **structure, readability, and clarity** of the report in presenting results are important.
- Students must clearly communicate **what has been modelled, how the system works, and what assumptions were made.**
- Students must clearly state the **aim of the project** in the introduction, and this must be followed up and answered throughout the report.
- Students must be able to **analyse and discuss simulation results, and draw conclusions regarding energy savings, profitability, and climate impact**.
- Students must be able to **compare different energy measures** (heat pumps, solar PV, wind, storage, insulation improvements, etc.) and evaluate how effective the solutions are in different conditions or locations.
- Students must demonstrate proper **use of sources** (literature, data, reports) and show **critical evaluation of them**.

Group Agreement and work plan (to be written together in the group)

The group must write a **collaboration agreement and a project plan**:

The project plan should focus on the practical implementation of the academic/technical part of the project. The collaboration agreement should address all the non-technical aspects that come with working in a group project (e.g., communication, responsibilities, meeting routines, conflict resolution).

5. Suggested Timeline

- **Week 1**: Goup agreement. Collect weather + house data, review libraries.
- **Week 2**: Build baseline house thermal model.

- **Week 3**: Add heating system + run baseline simulation.
- **Week 4-6**: Implement renewable and storage scenarios.
- **Week 7-9**: Results comparison and final report.
- **Week 10**: Presentation

6. Learning Outcomes

- Apply **Modelica/Dymola modeling** to real-world energy systems.
- Understand **building physics** (heat loss, capacity, comfort).
- Learn how **renewables and storage** interact with demand.
- Gain skills in **scenario analysis and sustainable design**.

Good luck !